Stark Shifts in Simulated Excited State Vibrational Spectra

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Abstract

Our chosen system of study is the *2-[(7-4-[N,N-Bis(4-methylphenyl)amino]phenyl-2,1,3- benzothiadiazol-4-yl)methylene]propanedinitrile* (DTDCPB) chromophore prevalent in organic photovoltaic materials research. DTDCPB has nitrile groups that make it ideal for IR studies due to its strong signature stretching frequency. For the first component of our project, we hope to simulate some ground state IR spectra of DTDCPB in the presence of varying external fields. This can then be used to calculate the Stark shift and build a calibration curve for the effect of external field on the Stark shift of this molecule. Furthermore, we plan to probe the excited state of DTDCPB as well and see if we can look at Stark effects in the excited state as well. The third step we hope to investigate is a pump-probe transient IR study of this molecule where the excited state dynamics of DTDCPB can be observed. Hopefully our understanding of Stark effects in the ground and excited states will elucidate some of the dynamics of the transient IR study.

Introduction

As research in solar energy becomes increasingly important for environmental applications, several new strategies are being investigated to increase the efficiency of these devices. Specificlly in organic photovoltaics (OPVs), one such strategy has been to use small molecules with large ground-state dipole moments. It is believed that these dipoles can allow for enhanced hole hopping and can minimize charge recombination (ref Griffith). One particularly promising molecule is 2-[(7-4-[N,N-Bis(4-methylphenyl)amino]phenyl-2,1,3-benzothiadiazol-4-yl)methylene] propanedinitrile (DTDCPB) and has resulted in OPVs with power conversion effiencies of up to 9.6%. The nitrile groups present in teh molecule can act as reporter moieties for vibrational spectroscopy. By modeling the effect of an applied electric field on absorption spectrum (Stark Effect), it is then possible to gather information regarding the local electric field during charge transfer events in a photovoltaic device via excited state IR spectroscopy. The first requires a thorough understanding of the electric field dependence of the ground state IR spectrum before probing the excited state.

Theory

Some theoretical hand-wavey bullshit and lots of "beautiful mathematics" shall go here. Someday.

Here is an inline expression: $N = \frac{1}{2}\hbar$. Whereas here we have a single line expression:

$$\Psi = \sum_i A_{ij} c^\dagger.$$

$$\begin{aligned} A + (B + C) &= (A + B) + C \\ A &= \sum k_{ij}^2 \end{aligned}$$

Methodology

The Stark effect comes in when a field is applied.

Applied fields to observe Stark shifts

Calibration curve:

Apply external E field

Measure IR spectrum

Calibrate Stark shifts for use in Pump Probe

Watch Charge transfer (local E field)

Easy to put in an external field (molecule comes into Equilibrium in the field) to see the IR

The molecule is geometry optimized, then frequency calculated

Static = x,y,z then value of the field

Geometry optimization in a static E field (google the warnings to explain it)

Include point charges (dummy charges)

Dipole along z-axis, put two point charges on the axis and vary the distance to vary magnitude of the field.

Start at ground state Calibration Curve thingy for Stark Shifts Static E fields

a. Opt+freq $\longrightarrow$ postprocess for IR

Excited state opt+freq $\longrightarrow$ postprocess for IR

a. Same process

Model pump probe using Stark shift Calibration (dynamics, no ext field)

a. BOMD $\longrightarrow$ freq postprocess for IR

?????? PROFIT!!!!

Computational Results

Discussion

Conclusion